

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12385550
Kind Code	B1
Date of Patent	August 12, 2025
Inventor(s)	Chung; Chi-Wen et al.

Cycloid speed reducer

Abstract

A cycloid speed reducer is disclosed. The accommodation space required for the sealing component is integrated into the track ring to save the material costs and significantly reduce the overall volume. Moreover, a spacer ring is placed between the front bearing and the rear bearing, and the thickness of the spacer ring can be used to adjust the clearances of the front bearing and the rear bearing to facilitate the quantitative production and control the yield. On the other hand, the initial shape of the output plate is adjusted through mass production molding, and the thickness next to the protruding structures of the output plate is increased, so that the supporting plane can be milled through a lower-cost lathe processing, and the accuracy of bearing plane can meet the requirements. There is no need to perform full-circle milling on the output plate, and the processing hours are reduced significantly.

Inventors:	Chung; Chi-Wen (Taoyuan, TW), Chu; Wei-Ying (Taoyuan, TW)
Applicant:	Delta Electronics, Inc. (Taoyuan, TW)
Family ID:	96661874
Assignee:	Delta Electronics, Inc. (Taoyuan, TW)
Appl. No.:	18/825898
Filed:	September 05, 2024

Foreign Application Priority Data

CN	202411127815.6	Aug. 16, 2024
----	----------------	---------------

Related U.S. Application Data

us-provisional-application US 63631617 20240409

Publication Classification

Int. Cl.: F16H1/32 (20060101); F16H57/021 (20120101)

U.S. Cl.:

CPC F16H1/32 (20130101); F16H57/021 (20130101); F16H2001/323 (20130101)

Field of Classification Search

CPC: F16H (1/32); F16H (2001/323); F16H (2001/327); F16H (57/029); F16H (2001/325)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
6492787	12/2001	Hibino et al.	N/A	N/A
7988581	12/2010	Kobayashi	N/A	N/A
9556933	12/2016	Fujimoto	N/A	N/A
9976629	12/2017	Shen	N/A	N/A
10673304	12/2019	Tsai et al.	N/A	N/A
10677321	12/2019	Tsai et al.	N/A	N/A
11486469	12/2021	Chung	N/A	F16H 1/32
11555531	12/2022	Chung	N/A	F16H 1/34
11841073	12/2022	Kihira	N/A	F16H 57/0471
2019/0207470	12/2018	Uematsu et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
103968009	12/2013	CN	F16H 1/32
105020345	12/2014	CN	F16H 1/32
107237866	12/2016	CN	N/A
107387677	12/2016	CN	F16H 1/32
108019471	12/2017	CN	F16H 1/32
110185748	12/2018	CN	B25J 17/00
111853167	12/2019	CN	F16H 1/32
114251418	12/2021	CN	N/A
115126845	12/2021	CN	N/A
115431254	12/2021	CN	N/A
115461968	12/2021	CN	N/A
116221345	12/2022	CN	N/A
117722482	12/2023	CN	N/A
117869543	12/2023	CN	F16H 1/32
102017126737	12/2018	DE	F16H 1/32
2579204	12/2019	GB	N/A
201627589	12/2015	TW	N/A
202200330	12/2021	TW	N/A
202407236	12/2023	TW	N/A

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application claims the benefit of U.S. Provisional Application No. 63/631,617 filed on Apr. 9, 2024, and entitled “CYCLOID SPEED REDUCER ASSEMBLY”. This application claims priority to China Patent Application No. 202411127815.6, filed on Aug. 16, 2024. The entireties of the above-mentioned patent applications are incorporated herein by reference for all purposes.

FIELD OF THE INVENTION

(1) The present disclosure relates to a speed reducer, and more particularly to a cycloid speed reducer optimizing the component structure to realize the miniaturization application.

BACKGROUND OF THE INVENTION

(2) Currently on the market, the speed reducers commonly used in robotic arms can be roughly divided into two structural types: “cycloid speed reducer” and “harmonic speed reducer”. Both of these reducers are lightweight, compact and have high reduction ratio characteristics.

(3) The harmonic speed reducer includes a wave generator, a flexible gear and a rigid gear. The elastic deformation of the flexible gear can be controlled to result in a pushing action so as to transmit motion and power of mechanical transmission. The harmonic speed reducer has smaller size, lighter weight and higher precision when compared with the RV reducer. However, since the rigidity of the flexible gear of the harmonic speed reducer is low, the harmonic speed reducer is unable to withstand high impact and has a problem of causing teeth difference friction. In other words, the use life of the harmonic speed reducer is shorter.

(4) Conventionally, the cycloid speed reducer includes an eccentric shaft and two cycloid discs. Each of the two cycloid discs includes at least one tooth. Moreover, the two cycloid discs are linked with a power input shaft and a power output shaft, respectively. During operations of the cycloid speed reducer, one cycloid disc is rotated with the power input shaft through the eccentric shaft, and the power output shaft is rotated with the other cycloid disc. Through the corresponding teeth, the two cycloid discs are correspondingly rotated. The conventional cycloid speed reducer has many benefits such as high transmission ratio, compact structure and high transmission efficiency. However, compared with the harmonic speed reducer, the conventional cycloid speed reducer still has a larger size and is not conducive to miniaturization.

(5) Therefore, there is a need of providing a cycloid speed reducer optimizing the component structure to realize the miniaturization application, so as to overcome the above drawbacks.

SUMMARY OF THE INVENTION

(6) An object of the present disclosure is to provide a cycloid speed reducer optimizing the component structure to realize the miniaturization application. In order to achieve the sealing design of the front end and the rear end of the cycloid speed reducer of the present disclosure, the accommodation space required for the sealing element is integrated into the track ring, and the traditional end cover design is omitted to save the material costs and significantly reduce the overall volume of the cycloid speed reducer. Moreover, in order to prevent the front bearing and the rear bearing of the input shaft from affecting the transmission due to installation clearance errors, a spacer ring is placed between the front bearing and the rear bearing, and the thickness of the spacer ring can be used to adjust the clearances of the front bearing and the rear bearing to facilitate the quantitative production and control the yield. On the other hand, the position-limitation of the gear plate in the cycloid reducer usually relies on the output plate or the track ring to act as the

supporting and limiting parts. In the present disclosure, the initial shape of the output plate is adjusted through mass production molding, and the thickness next to the protruding structures of the output plate is increased, so that the supporting planes can be milled through a lower-cost lathe processing, and the accuracy of supporting plane can meet the requirements. There is no need to perform full-circle milling on the output plate, and the processing hours are reduced significantly. In this way, by optimizing the aforementioned component structure, the cycloid speed reducer of the present disclosure not only achieves miniaturization application, but also saves the material and manufacturing costs.

(7) In accordance with an aspect of the present disclosure, a cycloid speed reducer is provided, and includes an input shaft, a cycloidal gear plate, a roller wheel assembly, an output plate, a track base and a sealing element. The input shaft is arranged along an axial direction. The cycloidal gear plate includes a shaft hole and an outer tooth portion, wherein the shaft hole passes through the cycloidal gear plate along the axial direction, and is configured to be passed through by the input shaft, and the outer tooth portion is disposed on an outer periphery of the cycloidal gear plate. The roller wheel assembly is sleeved on the cycloidal gear plate and includes a plurality of rollers spatially corresponding to the outer tooth portion of the cycloidal gear plate. The output plate is connected to the cycloidal gear plate through an eccentric transmission shaft, wherein when the input shaft drives the outer tooth portion of the cycloidal gear plate to engage with the corresponding rollers of the roller wheel assembly, the cycloid gear plate is rotated with the eccentric transmission shaft, so that the output plate is driven by the eccentric transmission shaft to rotate. The track base is disposed on an outer side of the roller wheel assembly, and spatially corresponding to an outer peripheral wall of the output plate, wherein the track base is extended outward along the outer peripheral wall of the output plate to form an accommodation space between the track base and the outer peripheral wall of the output plate. The sealing element is received within the accommodation space and tightly fitted between the track base and the output plate.

(8) In an embodiment, the cycloid speed reducer further includes a roller bearing, wherein the roller bearing is arranged between the output plate and the track base, and includes a plurality of rollers configured to run between the track base and the output plate.

(9) In an embodiment, the track base is spatially corresponding to the output plate, and a parallelogram is collaboratively formed on a radial section so that the plurality of rollers of the bearing roller are configured to run between the track base and the output plate.

(10) In an embodiment, the track base includes a groove, which is ring-shaped on an inner peripheral wall of the track base and configured to engage with an outer peripheral edge of the sealing element.

(11) In an embodiment, the cycloid speed reducer further includes a deep groove bearing, a needle bearing and a spacer ring, wherein the input shaft includes a deep-groove-bearing connection section and a needle-bearing connection section, the deep groove bearing is connected between the deep-groove-bearing connection section and the output plate, the needle bearing is connected between the needle-bearing connection section and the cycloidal gear plate, and the deep groove bearing and the needle-bearing connection section are connected through the spacer ring.

(12) In an embodiment, the spacer ring includes a first spacer section and a second spacer section connected to each other along the axial direction to form a stepped structure.

(13) In an embodiment, an inner ring of the deep groove bearing interferes with the first spacer section for bearing, and the second spacer section interferes with the needle-bearing connection section for bearing.

(14) In an embodiment, the first spacer section has a first spaced outer diameter, the second spacer section has a second spaced outer diameter, and the first spaced outer diameter is less than the second spaced outer diameter.

(15) In an embodiment, the deep-groove-bearing connection section has a deep-groove-bearing inner diameter, the needle-bearing connection section connected to the needle bearing has a needle-

bearing inner diameter, and an eccentricity value is formed by the relative distance between a central axis of the needle-bearing connection section and a central axis of the deep-groove-bearing connection section, wherein the first spaced outer diameter is greater than the sum of the deep-groove-bearing inner diameter and 1.5 mm, and is less than or equal to the needle-bearing inner diameter, wherein the second spaced outer diameter is greater than the sum of the first spaced outer diameter and 2 times the eccentricity value, and is less than or equal to the sum of the needle-bearing inner diameter, the difference between the needle-bearing inner diameter and the deep-groove-bearing inner diameter and 2 times the eccentricity value.

(16) In an embodiment, the output plate includes a plurality of protruding structures and a plurality of supporting planes, the plurality of protruding structure are disposed on a datum plane, and the plurality of supporting planes are correspondingly connected to the plurality of protruding structures and higher than the datum plane.

(17) In an embodiment, the plurality of supporting planes are formed after a plurality of milling outreach structures for increasing thickness of the plurality of protruding structures are removed through milling, and the plurality of supporting planes are spaced apart from each other and configured to support a bottom surface of the cycloidal gear plate.

(18) In an embodiment, the output plate includes a first output plate and a second output plate, the first output plate and the second output plate are respectively located on two opposite outer sides of the roller wheel assembly, and each provides a power output, respectively.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The above contents of the present disclosure will become more readily apparent to those ordinarily skilled in the art after reviewing the following detailed description and accompanying drawings, in which:

(2) FIG. 1 is a perspective view illustrating a cycloid speed reducer according to an embodiment of the present disclosure;

(3) FIG. 2 is a cross-sectional view illustrating the cycloid speed reducer according to the embodiment of the present disclosure;

(4) FIG. 3 is a cross-sectional structural view illustrating the portion of the cycloid speed reducer adjacent to the track bases according to the embodiment of the present disclosure;

(5) FIG. 4 is a cross-sectional view illustrating the dimensional relationship between the bearing, the spacer ring and the input shaft in the cycloid speed reducer of the embodiment of the present disclosure;

(6) FIG. 5 is a perspective view illustrating the spacer ring of the cycloid speed reducer according to the first embodiment of the present disclosure;

(7) FIG. 6 is a perspective view illustrating an initial shape of the first output plate in the cycloid speed reducer before milling according to the first embodiment of the present disclosure;

(8) FIG. 7 is a perspective view illustrating the shape of the first output plate in the cycloid speed reducer after milling according to the first embodiment of the present disclosure; and

(9) FIG. 8 is a cross-sectional structural view illustrating the first output plate supporting the cycloid gear plate in the cycloid speed reducer according to the first embodiment of the present disclosure.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENT

(10) The present disclosure will now be described more specifically with reference to the following embodiments. It is to be noted that the following descriptions of preferred embodiments of this disclosure are presented herein for purpose of illustration and description only. It is not intended to be exhaustive or to be limited to the precise form disclosed. For example, the formation of a first

feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed. Further, spatially relative terms, such as “front,” “rear” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly. When an element is referred to as being “connected,” or “coupled,” to another element, it can be directly connected or coupled to the other element or intervening elements may be present. Although the wide numerical ranges and parameters of the present disclosure are approximations, numerical values are set forth in the specific examples as precisely as possible. In addition, although the “first,” “second,” and the like terms in the claims be used to describe the various elements can be appreciated, these elements should not be limited by these terms, and these elements are described in the respective embodiments are used to express the different reference numerals, these terms are only used to distinguish one element from another element. For example, a first element could be termed a second element, and, similarly, a second element could be termed a first element, without departing from the scope of example embodiments. Besides, “and/or” and the like may be used herein for including any or all combinations of one or more of the associated listed items.

(11) FIG. 1 is a perspective view illustrating a cycloid speed reducer according to an embodiment of the present disclosure. FIG. 2 is a cross-sectional view illustrating the cycloid speed reducer according to the first embodiment of the present disclosure. Please refer to FIG. 1 and FIG. 2. The present disclosure provides a cycloid speed reducer **1**. Preferably but not exclusively, the cycloid speed reducer **1** can be applied to motors, machine tools, robotic arms, automobiles, motorcycles or other power assisting devices in order to provide a speed reducing function. In the embodiment, the cycloid speed reducer **1** includes an input shaft **10**, a cycloidal gear plate **11**, an eccentric transmission shaft **13**, a first output plate **14a**, a second output plate **14b** and a roller wheel assembly **12**. The roller wheel assembly **12** includes a wheel disc **121** and a plurality of rollers **122**. The wheel disc **121** includes a shaft hole (not shown), which is located at the substantial center position of the wheel disc **121** for a part of the input shaft **10** to pass through. Moreover, the wheel disc **121** is driven by the input shaft **10** to rotate. The plurality of rollers **122** are disposed on the wheel disc **121**. The input shaft **10** is arranged along an axial direction C. Preferably but not exclusively, the input shaft **10** is configured to receive a power input provided by, for example, a motor (not shown), and is driven to rotate by the power input. The cycloidal gear plate **11** includes a shaft hole **110** and an outer tooth portion **111**. The shaft hole **110** is located at the substantial center position of the cycloidal gear plate **11** and spatially corresponding to the input shaft **10** in arrangement position. The shaft hole **110** is passed through by a part of the input shaft **10**, so that the cycloidal gear plate **11** is sleeved on the input shaft **10**. When the input shaft **10** is rotated, the cycloidal gear plate **11** is driven by the input shaft **10** to rotate. The roller wheel assembly **12** is sleeved on the cycloidal gear plate **11** and includes the plurality of rollers **122** spatially corresponding to the outer tooth portion **111** of the cycloidal gear plate **11**. Preferably but not exclusively, the outer tooth portion **111** can be protruded from an outer periphery of the cycloidal gear plate **11**. Moreover, the outer tooth portion **111** of the cycloidal gear plate **11** is in contact with the corresponding rollers **122** of the roller wheel assembly **12**. When the power input provided by the motor is inputted through the input shaft **10**, the outer tooth portion **111** of the cycloidal gear

plate **11** is driven directly to engage with the corresponding rollers **122** of the roller wheel assembly **12**.

(12) In the embodiment, the cycloidal gear plate **11** is connected to the input shaft **10** through a needle bearing **20**, and connected to the eccentric transmission shaft **13** through a needle bearing **21**, respectively. Preferably but not exclusively, the eccentric transmission shaft **13** is a crankshaft including four eccentric sections with the same diameter, and there are five eccentric transmission shaft **13** in the cycloid speed reducer **1**. Each transmission shaft **13** includes two concentric ends disposed at two opposite ends thereof and two eccentric ends arranged between the two concentric ends. The two eccentric ends are spatially corresponding to the cycloidal gear plate **11**. When the cycloid gear plate **11** is driven by the input shaft **10** to rotate, the cycloidal gear plate **11** drives the eccentric transmission shaft **13** to rotate by connecting with the eccentric ends of the eccentric transmission shaft **13**. In that, the concentric ends of the eccentric transmission shaft **13** are rotated synchronously, and the first output plate **14a** and the second output plate **14b** are driven by the concentric ends of the eccentric transmission shaft **13** to rotate respectively. The first output plate **14a** and the second output plate **14b** are located at two opposite outer ends of the cycloid speed reducer **1**. Preferably but not exclusively, the first output plate **14a** is disposed adjacent to the motor and can be regarded as the rear end of the cycloid speed reducer **1**, and the second output plate **14b** is disposed at another opposite end and regarded as the front end of the cycloid speed reducer **1**. Certainly, the present disclosure is not limited thereto. In the embodiment, at least one of the first output plate **14a** and the second output plate **14b** can be used as the power output of the cycloid speed reducer **1**.

(13) In the embodiment, the first output plate **14a** and the second output plate **14b** are located at two opposite sides of the roller wheel assembly **12**. In that, the cycloidal gear plate **11** is located between the first output plate **14a** and the second output plate **14b**. Preferably but not exclusively, both of the first output plate **14a** and the second output plate **14b** are served as the output end of the cycloid speed reducer **1** to provide the power output. In the embodiment, there are two sets of roller bearings **15**, one set is arranged between the first output plate **14a** and one side of the roller wheel assembly **12**, and another set is arranged between the second output plate **14b** and another opposite side of the roller wheel assembly **12**. Two track bases **17** are disposed on two opposite sides of the wheel disc **121** of the roller wheel assembly **12**, respectively. The two track bases **17** are spatially corresponding to the first output plate **14a** and the second output plate **14b**, respectively, and a parallelogram is collaboratively formed on a radial section so that the plurality of rollers in two sets of bearing rollers **15** are configured to run between the track base **17** and the first output plate **14a** and between the track base **17** and the second output plate **14b**.

(14) Notably, in order to further realize the miniaturization, the design of the cycloid speed reducer **1** in the present disclosure further omits the conventional front-end and rear-end cover structures for sealing grooves. FIG. **3** is a cross-sectional structural view illustrating the portion of the cycloid speed reducer adjacent to the track bases according to the embodiment of the present disclosure. Taking the second output plate **14b** as an example, the track base **17** is further extended outward along the outer peripheral wall of the second output plate **14b** to form an accommodation space **170**, which is configured to receive a sealing element **18**. Through the accommodation space **170** formed by the track base **17**, the sealing element **18** is tightly fitted between the track base **17** and the second output plate **14b** to achieve the front end sealing of the cycloid speed reducer **1**. Similarly, the track base **17** and the sealing element **18** mentioned above can be disposed correspondingly to the first output plate **14a**, so that the rear end sealing of the cycloid speed reducer **1** is realized. Certainly, the manner in which the sealing element **18** is tightly fitted between the track base **17** and the first output plate **14a** or between the track base **17** and the second output plate **14b** can be adjusted according to the practical requirements. Preferably but not exclusively, the sealing element **18** is a sealing ring structure. The track base **17** further includes a groove **171** spatially corresponding to the accommodation space **170**. The groove **171** is ring-shaped on an

inner peripheral wall of the track base **17** and configured to engage with an outer peripheral edge of the sealing element **18**. Certainly, the present disclosure is not limited thereto. Compared with the traditional sealing method of covering the ring teeth disk of the reducer with the front end cover and rear end cover installed, the accommodation space **170** for the seal element **18** is integrated into the track base **17**, and the traditional end cover design can be omitted. It has higher flexibility in application. Certainly, the present disclosure is not limited thereto.

(15) In the embodiment, the input shaft **10** is directly driven by the power source input from the motor (not shown), so that the cycloidal gear plate **11** and the roller wheel assembly **12** are interacted with each other. Therefore, the stability of the input source is extremely important. In order to ensure the stability of the input shaft **10**, the arrangement of deep groove bearings **16** is used to eliminate backlash and generate pre-pressure in the present disclosure. In the embodiment, the input shaft **10** and the first output plate **14a** or/and the second output plate **14b** are connected through the deep groove bearings **16**. The cooperation between the shaft hole of the deep groove bearings **16** and the input shaft **10** is mainly an interference fit, so as to ensure the stability of the input shaft **10**. However, the deep groove bearings **16** typically have large tolerances, and the accumulation of excessive tolerance values will not ensure that the assembled deep groove bearings **16** can generate overpressure to achieve the effects of eliminating backlash or generating pre-pressure. Therefore, in the present disclosure, a thickness of a spacer ring **19** is used to adjust the clearances, so as to solve the cumulative tolerance problem that is difficult to control in the bearings.

(16) FIG. 4 is a cross-sectional view illustrating the dimensional relationship between the bearing, the spacer ring and the input shaft in the cycloid speed reducer of the embodiment of the present disclosure. FIG. 5 is a perspective view illustrating the spacer ring of the cycloid speed reducer according to the embodiment of the present disclosure. Please refer to FIG. 1 to FIG. 5. In the embodiment, the input shaft **10** includes a deep-groove-bearing connection section **101** and a needle-bearing connection section **102**. The deep groove bearings **16** are connected the deep-groove-bearing connection section **101** and the first output plate **14a**/the second output plate **14b**. The needle bearing **20** is connected between the needle-bearing connection section **102** and the cycloidal gear plate **11**. In the embodiment, the spacer ring **19** includes a first spacer section **191** and a second spacer section **192**, which are connected to each other along the axial direction C to form a stepped structure. An outer rings of the bearings **16** interfere with the first output plate **14a** and the second output plate **14b**, respectively. An inner ring of the deep groove bearing **16** interferes with the first spacer section **191** of the spacer ring **19** for bearing, and the second spacer section **192** of the spacer ring **19** interferes with the needle-bearing connection section **102** of the input shaft **10** for bearing. In the embodiment, the first spacer section **191** has a first spaced outer diameter $\varnothing d$ that is less than a second spaced outer diameter $\varnothing D$ of the second spacer section **192**. In the embodiment, the deep-groove-bearing connection section **101** connected to the deep groove bearing **16**, and has a deep-groove-bearing inner diameter $\varnothing A$. The needle-bearing connection section **102** connected to the needle bearing **20** has a needle-bearing inner diameter $\varnothing B$, and an eccentricity value e is formed by the relative distance between the central axis of the needle-bearing connection section **102** and the central axis of the deep-groove-bearing connection section **101**. In the embodiment, in order to ensure that the assembled deep groove bearings **16** can generate overpressure to achieve the effect of eliminating backlash or generating pre-pressure, the size of the spacer ring **19** is further designed to match the size of the input shaft **10**. In the embodiment, the first spaced outer diameter $\varnothing d$ is greater than the sum of the deep-groove-bearing inner diameter $\varnothing A$ and a processing thickness. Preferably but not exclusively, the processing size is measured in mm, and the processing thickness can be for example 1.5 mm. Moreover, the first spaced outer diameter $\varnothing A$ is less than or equal to the needle-bearing inner diameter $\varnothing B$, as shown in the following equation (1). In the embodiment, the second spaced outer diameter $\varnothing D$ is greater than the sum of the first spaced outer diameter $\varnothing d$ and 2 times the eccentricity value e , and is less

than or equal to the sum of the needle-bearing inner diameter $\varnothing B$, the difference between the needle-bearing inner diameter $\varnothing B$ and the deep-groove-bearing inner diameter $\varnothing A$ and 2 times the eccentricity value e , as shown in the following equation (2).

$$\varnothing A + \text{processing thickness} < \varnothing d \leq \varnothing B \quad (1)$$

$$\varnothing d + 2e < \varnothing D \leq \varnothing B + (\varnothing B - \varnothing A) + 2e \quad (2)$$

(17) Certainly, the size design of the spacer ring **19** is adjustable according to the sizes of the deep groove bearings **16**, the first output plate **14a** or the second output plate **14b** or the practical requirements, and the present disclosure is not limited thereto. In addition, notably, since the deep groove bearings **16** of the present disclosure are disposed inside the reducer, compared to the conventional structure in which the bearing is arranged on an aluminum cover, the structure of the input shaft **10** of the cycloid speed reducer **1** in the present disclosure has higher stability. The cycloid speed reducer **1** of the present disclosure will not have problems such as thermal expansion and interference forces caused by the bearing connected to the aluminum cover.

(18) On the other hand, some parts of the cycloid speed reducer **1** have processing cost issues. In addition to high-precision requirements, the structure of some parts is also relatively complex. FIG. **6** is a perspective view illustrating an initial shape of the first output plate in the cycloid speed reducer before milling according to the embodiment of the present disclosure. FIG. **7** is a perspective view illustrating the shape of the first output plate in the cycloid speed reducer after milling according to the embodiment of the present disclosure. FIG. **8** is a cross-sectional structural view illustrating the first output plate supporting the cycloid gear plate in the cycloid speed reducer according to the embodiment of the present disclosure. In the embodiment, the first output plate **14a** is usually a disk-shaped workpiece with several protruding structures. In mass production of such special-shaped workpiece, production molding strategies can only be adopted to reduce costs. Furthermore, after production molding, it is necessary to control the dimensional accuracy through machine processing. In the traditional machine processing method, too many surfaces need to be processed. The benefits of production molding cannot be increased. The position-limitation of the cycloidal gear plate **11** in the cycloid reducer **1** usually relies on for example the first output plate **14a** to act as the supporting and limiting parts. For the conventional output plate, the detailed processing and milling are necessary after production molding to ensure the accuracy of the bearing surface. The milling region needs to surround the entire circle of the output plate to ensure accuracy, so that a heavy cost is required. In the embodiment, the raw material **140** for the first output plate **14a** is shown in FIG. **6**. The protruding structures **142** are further respectively connected to milling outreach structures **143** and formed on the datum plane **141** to increase the thickness of the protruding structures **142**. The raw material **140** can be produced by molding. Thereafter, the fine structure of the first output plate **14a**, as shown in FIG. **7**, can be obtained through low-cost lathe processing. In the embodiment, the processing thickness of the protruding structures **142** is increased through the milling outreach structures **143**. After the milling outreach structures **143** are removed by milling, a plurality of supporting planes **145** are formed for the cycloidal gear plate **11**. The supporting planes **145** are higher than a datum plane **141** and spatially corresponding to the protruding structures **144** connected therewith. The plurality of supporting planes **145** are spaced apart from each other. In this way, the bottom surface **112** of the cycloidal gear plate **11** is accurately supported on the supporting planes **145**, as shown in FIG. **8**. Compared with the conventional processing method of milling the entire surface, the thickness next to the protruding structures **142** in the present disclosure is increased respectively with the milling outreach structures **143** through mass production molding. It avoids the problem caused by full-circle milling, and the processing hours are reduced significantly. In this way, by optimizing the aforementioned component structure, the cycloid speed reducer **1** of the present disclosure not only achieves miniaturization application, but also saves the material and manufacturing costs. Certainly, the present disclosure is not limited thereto.

(19) From the above, it can be seen that the cycloidal speed reducer **1** of the present disclosure

utilizes the design of the front-end and rear-end sealing element **18** and omits the conventional end cover design. It helps to save the material costs and significantly reduce the overall volume of the cycloid speed reducer **1**. Furthermore, with the spacer ring **19** placed between the front bearing and the rear bearing, it allows to prevent the front bearing and the rear bearing of the input shaft **10** from affecting the transmission due to installation clearance errors, and the thickness of the spacer ring **19** can be used to adjust the clearances of the front bearing and the rear bearing to facilitate the quantitative production and control the yield. Since the cycloid speed reducer **1** of the present disclosure provides the front output end and the rear output end at two opposite ends, it is more conducive to the assembly and application of miniaturized structures. For example, when the rear output end of the cycloid speed reducer **1** of the present disclosure is assembled with the motor, it facilitates to arrange an encoder between the speed reducer and the motor for closed-loop feedback control, so as to improve the control accuracy of the system. The application of the cycloid speed reducer assembly equipped with encoder feedback is generally applicable to general standard servo motors, and is competitive in terms of cost and convenience. Certainly, the applications of the cycloid speed reducer **1** are not limited thereto, and not redundantly described hereafter.

(20) In summary, the present disclosure provides a cycloid speed reducer optimizing the component structure to realize the miniaturization application. In order to achieve the sealing design of the front end and the rear end of the cycloid speed reducer of the present disclosure, the accommodation space required for the sealing element is integrated into the track ring, and the traditional end cover design is omitted to save the material costs and significantly reduce the overall volume of the cycloid speed reducer. Moreover, in order to prevent the front bearing and the rear bearing of the input shaft from affecting the transmission due to installation clearance errors, a spacer ring is placed between the front bearing and the rear bearing, and the thickness of the spacer ring can be used to adjust the clearances of the front bearing and the rear bearing to facilitate the quantitative production and control the yield. On the other hand, the position-limitation of the gear plate in the cycloid reducer usually relies on the output plate or the track ring to act as the supporting and limiting parts. In the present disclosure, the initial shape of the output plate is adjusted through mass production molding, and the thickness next to the protruding structures of the output plate is increased, so that the supporting planes can be milled through a lower-cost lathe processing, and the accuracy of supporting plane can meet the requirements. There is no need to perform full-circle milling on the output plate, and the processing hours are reduced significantly. In this way, by optimizing the aforementioned component structure, the cycloid speed reducer of the present disclosure not only achieves miniaturization application, but also saves the material and manufacturing costs.

(21) While the disclosure has been described in terms of what is presently considered to be the most practical and preferred embodiments, it is intended to cover various modifications and similar arrangements included within the spirit and scope of the appended claims which are to be accorded with the broadest interpretation so as to encompass all such modifications and similar structures.

Claims

1. A cycloid speed reducer, comprising: an input shaft arranged along an axial direction; a cycloidal gear plate comprising a shaft hole and an outer tooth portion, wherein the shaft hole passes through the cycloidal gear plate along the axial direction, and is configured to be passed through by the input shaft, and the outer tooth portion is disposed on an outer periphery of the cycloidal gear plate; a roller wheel assembly sleeved on the cycloidal gear plate and comprising a plurality of rollers spatially corresponding to the outer tooth portion of the cycloidal gear plate; an output plate connected to the cycloidal gear plate through an eccentric transmission shaft, wherein when the input shaft drives the outer tooth portion of the cycloidal gear plate to engage with the corresponding rollers of the roller wheel assembly, the cycloid gear plate is rotated with the

eccentric transmission shaft, so that the output plate is driven by the eccentric transmission shaft to rotate; a track base disposed on an outer side of the roller wheel assembly, and spatially corresponding to an outer peripheral wall of the output plate, wherein the track base is extended outward along the outer peripheral wall of the output plate to form an accommodation space between the track base and the outer peripheral wall of the output plate; and a sealing element received within the accommodation space and tightly fitted between the track base and the output plate, wherein the output plate comprises a plurality of protruding structures and a plurality of supporting planes, the plurality of protruding structures are disposed on a datum plane, and the plurality of supporting planes are correspondingly connected to the plurality of protruding structures and higher than the datum plane, wherein the plurality of supporting planes are formed after a plurality of milling outreach structures for increasing thickness of the plurality of protruding structures are removed through milling, and the plurality of supporting planes are spaced apart from each other and configured to support a bottom surface of the cycloidal gear plate.

2. The cycloid speed reducer according to claim 1, further comprising a roller bearing, wherein the roller bearing is arranged between the output plate and the track base, and comprises a plurality of rollers configured to run between the track base and the output plate.

3. The cycloid speed reducer according to claim 2, wherein the track base is spatially corresponding to the output plate, and a parallelogram is collaboratively formed on a radial section so that the plurality of rollers of the bearing roller are configured to run between the track base and the output plate.

4. The cycloid speed reducer according to claim 1, wherein the track base comprises a groove, which is ring-shaped on an inner peripheral wall of the track base and configured to engage with an outer peripheral edge of the sealing element.

5. The cycloid speed reducer according to claim 1, further comprising a deep groove bearing, a needle bearing and a spacer ring, wherein the input shaft includes a deep-groove-bearing connection section and a needle-bearing connection section, the deep groove bearing is connected between the deep-groove-bearing connection section and the output plate, the needle bearing is connected between the needle-bearing connection section and the cycloidal gear plate, and the deep groove bearing and the needle-bearing connection section are connected through the spacer ring.

6. The cycloid speed reducer according to claim 5, wherein the spacer ring comprises a first spacer section and a second spacer section connected to each other along the axial direction to form a stepped structure.

7. The cycloid speed reducer according to claim 6, wherein an inner ring of the deep groove bearing interferes with the first spacer section for bearing, and the second spacer section interferes with the needle-bearing connection section for bearing.

8. The cycloid speed reducer according to claim 6, wherein the first spacer section has a first spaced outer diameter, the second spacer section has a second spaced outer diameter, and the first spaced outer diameter is less than the second spaced outer diameter.

9. The cycloid speed reducer according to claim 8, wherein the deep-groove-bearing connection section has a deep-groove-bearing inner diameter, the needle-bearing connection section connected to the needle bearing has a needle-bearing inner diameter, and an eccentricity value is formed by the relative distance between a central axis of the needle-bearing connection section and a central axis of the deep-groove-bearing connection section, wherein the first spaced outer diameter is greater than the sum of the deep-groove-bearing inner diameter and 1.5 mm, and is less than or equal to the needle-bearing inner diameter, wherein the second spaced outer diameter is greater than the sum of the first spaced outer diameter and 2 times the eccentricity value, and is less than or equal to the sum of the needle-bearing inner diameter, the difference between the needle-bearing inner diameter and the deep-groove-bearing inner diameter and 2 times the eccentricity value.

10. The cycloid speed reducer according to claim 1, wherein the output plate comprises a first output plate and a second output plate, the first output plate and the second output plate are

respectively located on two opposite outer sides of the roller wheel assembly, and each provides a power output, respectively.
